FEVER: results tables

Table 1
Accuracy, precision, recall and F1 score (%) of the detectors at a 0.1 % false-alarm rate.

Detector	Attacks seen in training				Attacks seen only by another controller				New attack seen by no controller			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
FTOP	96.94	99.79	92.02	95.02	92.59	84.33	79.43	81.16	81.18	35.50	9.49	14.74
FTMaster	97.07	99.79	92.37	95.30	93.34	84.41	81.29	82.20	82.37	35.64	15.00	20.65
FloRa	99.01	**99.82**	97.50	98.62	93.65	84.30	82.19	82.68	84.51	35.48	24.90	28.46
DMS-BTT	97.03	99.78	92.27	95.22	93.45	84.29	81.62	82.38	82.21	35.36	14.28	19.94
FEVER	**99.61**	99.75	**99.19**	**99.46**	**99.43**	**99.75**	**98.60**	**99.15**	**89.48**	**98.99**	**50.98**	**66.81**

Same rows for every detector (simulator domains D1-D3).

Table 2
Comparison under LFTO attacks on the testbed (A2, A4 and E9P, five runs each).

Method	Benign flows harmed	Average attack share	Maximum attack share	First packets lost	New connections succeeding	Attackers blocked	Innocent hosts blocked
None	10.13 %	62.1 %	94.4 %	28.2 %	25.2 %	0 / 30	0
FTMaster	4.80 %	45.5 %	82.2 %	27.8 %	23.6 %	0 / 30	0
DMS-BTT	1.71 %	41.9 %	78.2 %	32.6 %	23.4 %	0 / 30	0
FEVER	**0.01 %**	**11.8 %**	**67.6 %**	**9.6 %**	**37.3 %**	30 / 30	0

Attack share: attack rules as a share of the core switch's table. FEVER with permanent blocking.

Table 3
Results under each attack.

Method	Benign flows harmed			Average attack share		
	A2	A4	E9P	A2	A4	E9P
None	10.29 %	6.03 %	14.08 %	58.4 %	54.8 %	73.1 %
FTMaster	**0.01 %**	4.16 %	10.22 %	10.0 %	53.4 %	73.3 %
DMS-BTT	0.02 %	2.09 %	3.01 %	18.7 %	47.6 %	59.3 %
FEVER	**0.01 %**	**0.02 %**	**0.00 %**	**5.7 %**	**16.4 %**	**13.4 %**

Table 4
Service to benign users under each attack.

Method	First packets lost			New connections succeeding		
	A2	A4	E9P	A2	A4	E9P
None	21.6 %	26.3 %	36.8 %	30.3 %	32.3 %	13.1 %
FTMaster	9.9 %	30.3 %	43.2 %	27.2 %	30.4 %	**13.3 %**
DMS-BTT	20.0 %	30.4 %	47.6 %	31.0 %	27.7 %	11.4 %
FEVER	**2.5 %**	**8.9 %**	**17.3 %**	**68.0 %**	**30.7 %**	13.2 %

Table 5
Benign traffic only (no attack, five runs).

Method	Benign flows harmed	First packets lost	New connections succeeding	Innocent hosts blocked
None	0.00 %	0.00 %	100.0 %	0
FTMaster	0.00 %	0.02 %	100.0 %	0
DMS-BTT	0.00 %	0.02 %	100.0 %	0
FEVER	0.00 %	0.17 %	100.0 %	0

Table 6
Effect of the block length (same runs).

Variant	Benign flows harmed			Average attack share	First packets lost	New connections succeeding
	A2	A4	E9P			
FEVER, 60 s block	1.05 %	0.32 %	0.10 %	21.9 %	19.8 %	29.8 %
FEVER, permanent block	**0.01 %**	**0.02 %**	**0.00 %**	**11.8 %**	**9.6 %**	**37.3 %**

Table 7
Comparison including rDefender (A2, A4 and E9P, three runs each).

Method	Benign flows harmed	Average attack share	First packets lost	New connections succeeding	Attackers blocked	Innocent hosts blocked
None	10.29 %	63.3 %	28.1 %	25.7 %	0 / 18	0
FTMaster	4.31 %	46.2 %	27.9 %	23.5 %	0 / 18	0
DMS-BTT	1.75 %	41.2 %	32.6 %	23.3 %	0 / 18	0
rDefender	1.83 %	45.0 %	36.4 %	20.8 %	0 / 18	0
FEVER	**0.02 %**	**11.1 %**	**9.9 %**	**33.0 %**	18 / 18	0

Seeds 4140-4142, the runs rDefender has.

Table 8
Ablation (A2, A4 and E9P, three runs each).

Variant	Benign flows harmed			Average attack share	Attackers blocked	Time to block under A2 (s)	Innocent hosts blocked
	A2	A4	E9P				
FEVER, 60 s block	0.61 %	**0.05 %**	0.16 %	**21.0 %**	18 / 18	**71**	0
Without volume blocking	**0.05 %**	1.11 %	3.89 %	43.4 %	6 / 18	**71**	0
Without detector	0.84 %	0.07 %	**0.00 %**	33.7 %	16 / 18	130	0

Run with the 60 s block, before permanent blocking; seeds 4140-4142.

Table 9
System overhead of each method on the testbed.

Method	CPU usage (%)			Memory (MB)		Time per decision (ms)
	No attack	Increase over no defense	Under attack	Peak	Increase over no defense	
None	48.2	-	92.9	216	-	30
FTMaster	48.6	+0.4	91.2	219	+2	72
DMS-BTT	52.4	+4.3	94.6	243	+27	65
FEVER	49.2	+1.1	88.9	291	+75	11